

Mathematics A

Unit 1 • Chapter OL-T17 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 11 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T17 • Expertise Q16+ • 3 items • 11 marks

Chapter	Items	Marks	Starts
01. OL-T17 Algebraic Roots & Indices	3	11	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebraic Roots & Indices

Unit 1 • Expertise • 11 marks • 3 item(s)

June 2024 | 4MA1-2H Q23 | 3 marks

Complex indexed monomial or fractional outer power

June 2022 | 4MA1-2H Q24 | 3 marks

Express one exponent variable in terms of others

January 2023 | 2H Q26 | 5 marks

Zero and negative indices

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q23

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

WORKING SPACE

4MA1-2H Q24 Q24

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

WORKING SPACE

26 Find the values of n such that

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

Show clear algebraic working.

(Total for Question 26 is 5 marks)

YOUR WORKING
